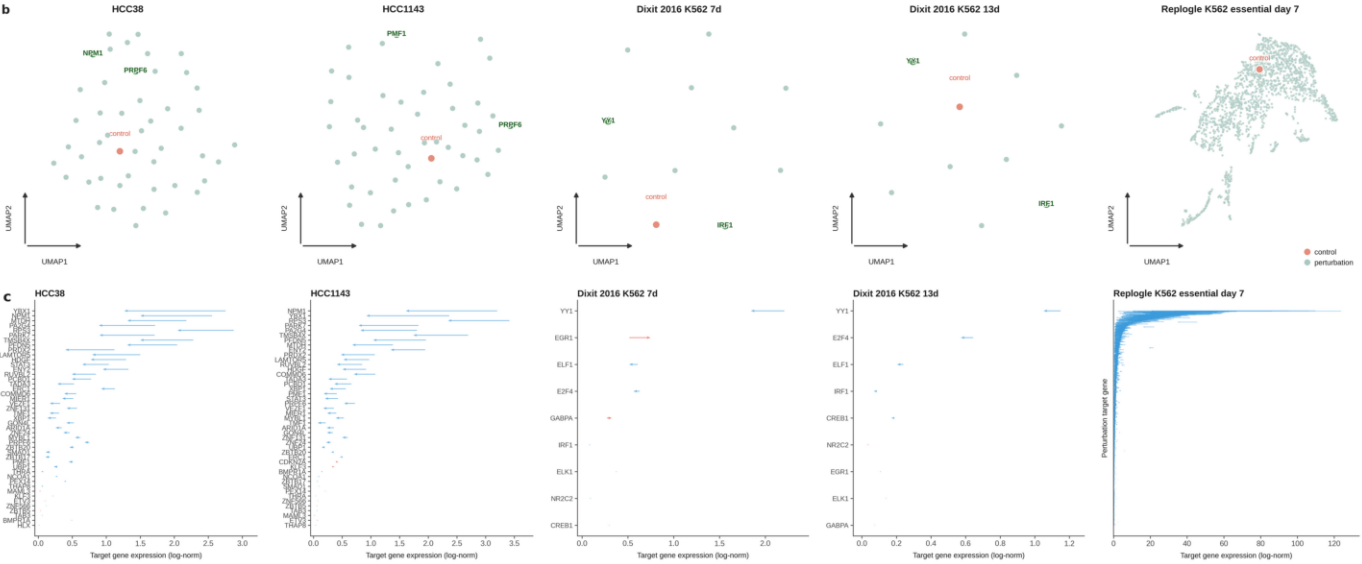
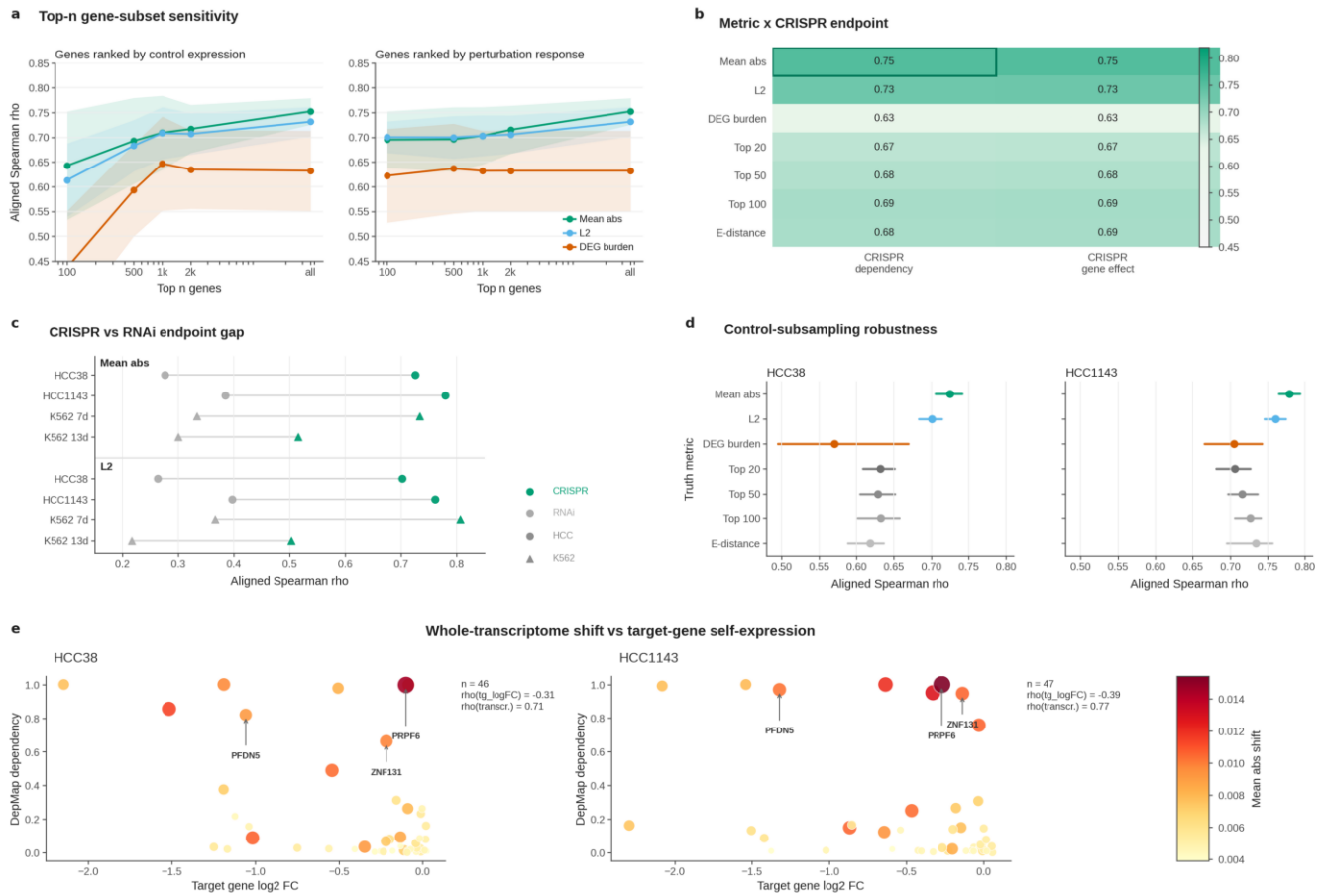


a

Dataset	Size	Benchmark use
Perturbation datasets		
HCC38	36,601 genes x 14,175 cells	47 targets 1,666 controls
HCC1143	36,601 genes x 11,405 cells	47 targets 1,325 controls
Dixit 2016 K562 7d	20,824 genes x 28,034 cells	10 targets 5,381 controls
Dixit 2016 K562 13d	20,824 genes x 15,849 cells	10 targets 3,491 controls
Replogle K562 essential day 7	8,563 genes x 310,385 cells	2057 targets 10,691 controls
Endpoint datasets		
DepMap CRISPR dependency (primary endpoint)	1,186 cell lines x 18,435 genes	HCC38 47 / HCC1143 48 / K562 7d 10 / K562 13d 10 / Replogle K562 1,882
DEMETER2 RNAi (sensitivity endpoint)	701 mapped cell lines x 17,309 genes	HCC38 48 / HCC1143 48 / K562 7d 9 / K562 13d 9 / Replogle K562 1,882

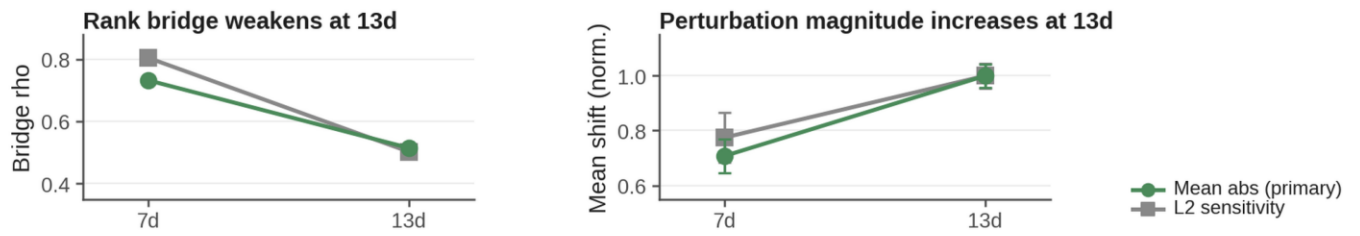


Extended Data Fig. 1 | Dataset familiarization. a, Dataset overview table. b, UMAP of perturbation-level mean profiles for HCC38, HCC1143, Dixit 2016 K562 7d, Dixit 2016. c, Target-gene expression change for the same five contexts (left to right).

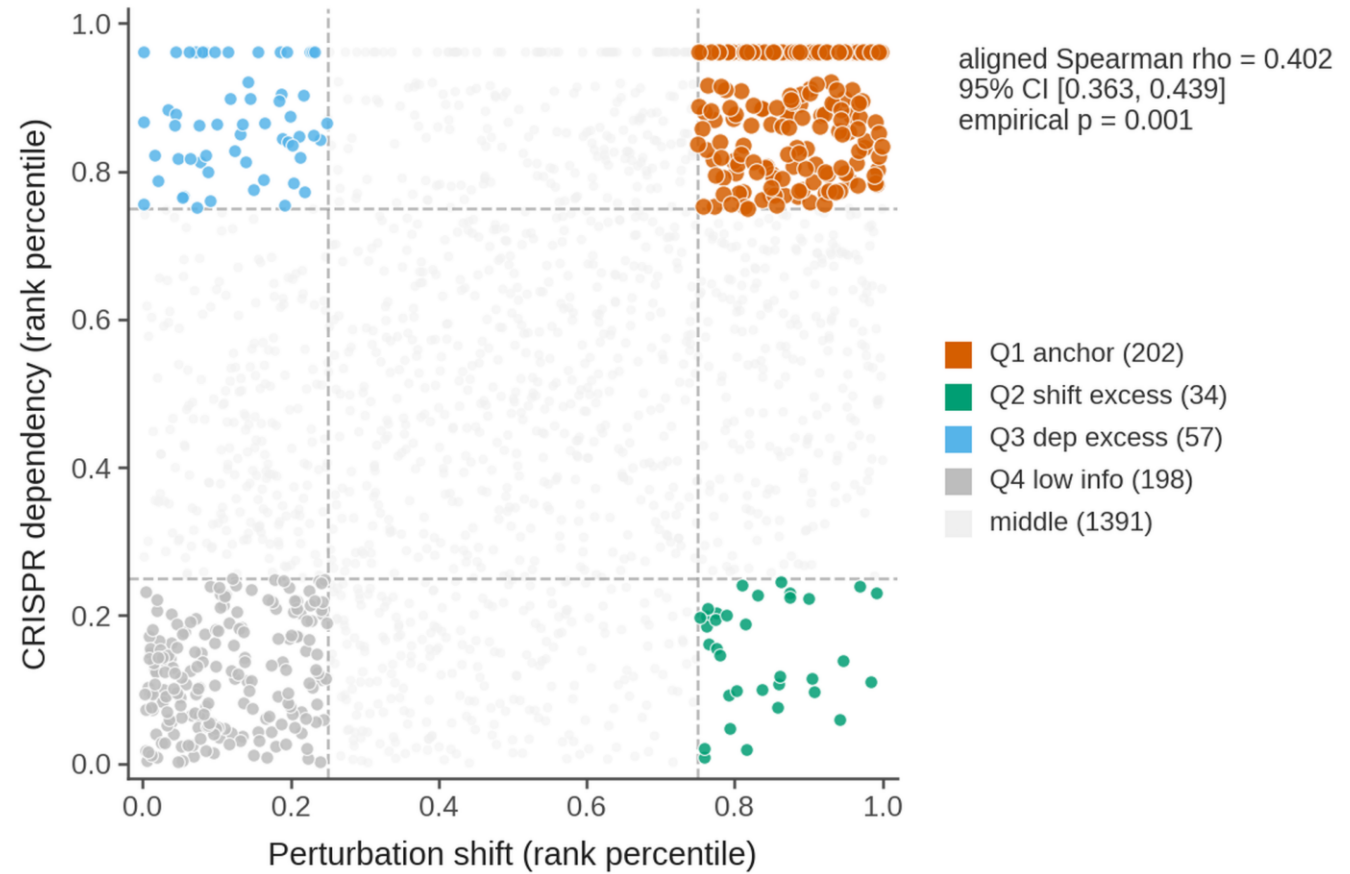


Extended Data Fig. 2 | Metric robustness. a, Top-n gene-subset sensitivity. b, Metric and CRISPR endpoint robustness. c, Endpoint sensitivity. d, Control-subsampling robustness. e, Whole-transcriptome shift versus target-gene self-expression in driving the fitness bridge.

a K562 temporal stratification (7d vs 13d)

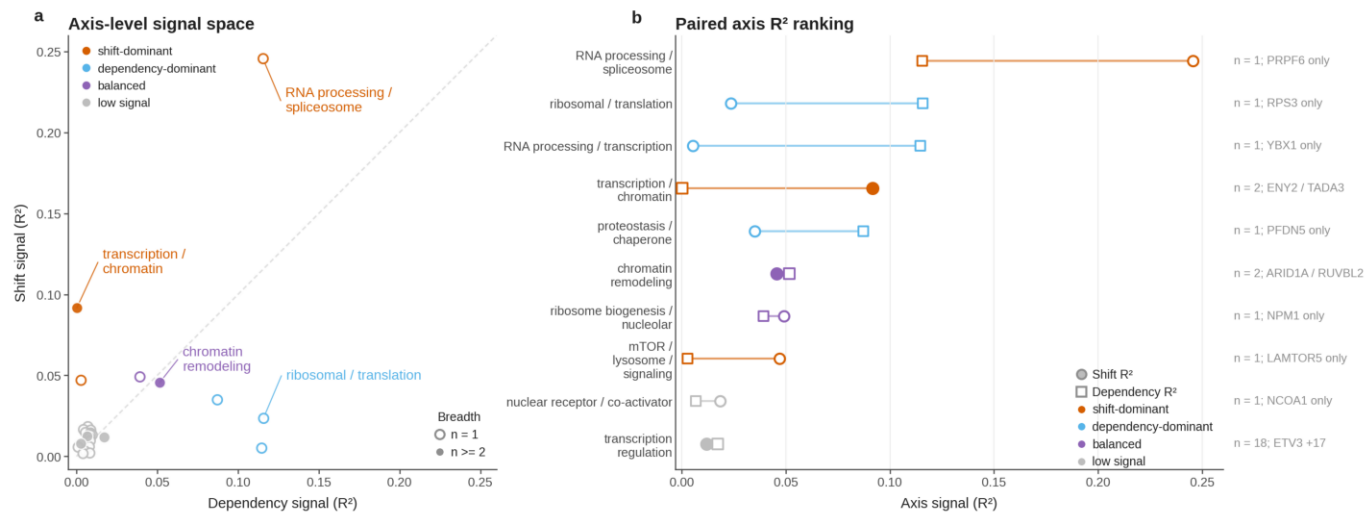


b Large-scale perturbation-fitness bridge confirmation



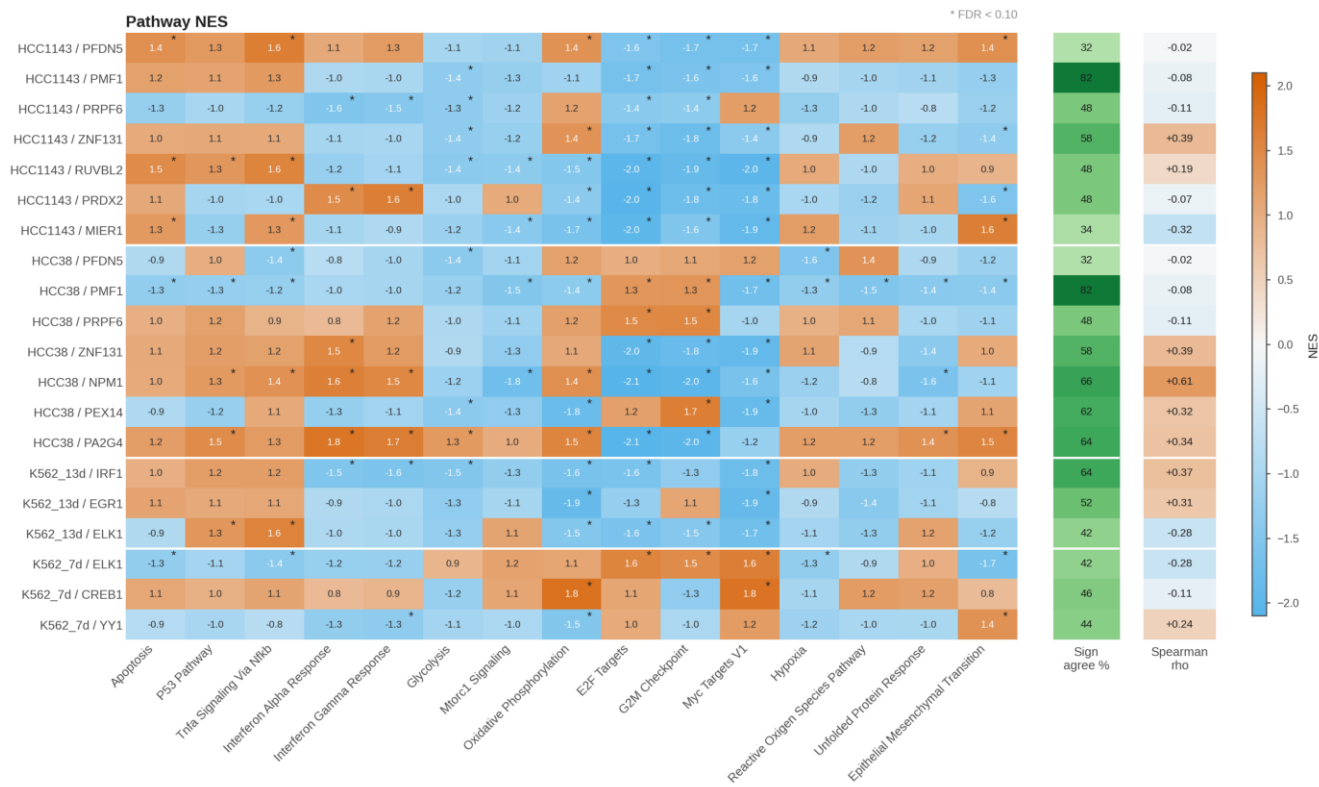
Extended Data Fig. 3 | K562 and Replogle bridge support.

a, K562 temporal bridge-magnitude dissociation (GSE90063). b, Replogle K562 essential day 7 large-scale perturbation-fitness bridge.



Extended Data Fig. 4 | Axis-level signal space. a, Axis-level signal space comparing dependency signal with transcriptomic shift signal across annotated axes. b, Paired axis R-squared ranking.

a Exploratory pathway-level summaries of target-control perturbation responses



Extended Data Fig. 5 | Pathway-response polarity. a, Pathway normalized enrichment score (NES) heat map across the pre-specified display targets in HCC38, HCC1143, K562 7d, and K562 13d.